Data Profiler

Data Preprocessing & Feature Engineering on Customer Churn Data

Project title	Data Profiler
Duration	5 hours
Author	Ahmed Bhesania
Dataset	customer_churn_200 (200 customers x 11 columns)
Target	Churn (Yes / No)
Notebook	main.ipynb (Parts A to E)
Profiling tool	YData Profiling v4.18.4 (report generated 28 Sep 2026)

1. Project Overview

The project takes a customer dataset through the full pre-machine-learning pipeline: acquiring data from four sources (CSV, JSON, MySQL and a REST API), understanding and cleaning it, exploring it with univariate, bivariate and multivariate analysis, and generating an automated profiling report. The end goal is to frame a supervised machine learning problem: **predict whether a customer will churn based on purchase behaviour**.

2. Tools and Libraries Used

Library / tool	Purpose in this project
Pandas	Reading CSV, JSON and SQL data; exploration (.head, .info, .describe); duplicate and missing-value handling; type conversion; string cleaning.
NumPy	Creating scalar, vector, matrix and 3-D tensor examples.
Matplotlib	Histograms, bar chart and figure layout for EDA.
Seaborn	KDE plots, box plots, correlation heatmap and pair plot.
mysql-connector-python	Connecting to the MySQL database `customer_churn` and reading the `customer_data` table.
Requests	Fetching JSON records from a REST API (Postman mock endpoint).
YData Profiling 4.18.4	Automated profiling report (Part E).
Jupyter Notebook	Environment for code, outputs and charts.

3. Dataset Description

Column	Type	Description
CustomerID	Text	Unique identifier (C001 to C200)
Age	Numeric	18 to 65 years
Gender	Categorical	Female (109) / Male (91) after cleaning
Income	Numeric	18,301 to 119,279
Purchases	Numeric	1 to 12
TotalSpent	Numeric	1,086 to 41,727
Visits	Numeric	2 to 16
SupportCalls	Numeric	0 to 6
City	Categorical	7 cities: Hyderabad 31, Ahmedabad 30, Surat 30, Bengaluru 29, Delhi 27, Pune 27, Mumbai 26
TenureMonths	Numeric	1 to 60 months
Churn	Target	No 122 (61%), Yes 78 (39%)

Descriptive statistics on the raw CSV (before imputation):

Variable	Count	Mean	Std	Min	Median	Max
Age	194	42.14	13.84	18	43	65
Income	195	70,696.35	30,163.81	18,301	72,340	119,279
Purchases	200	6.15	2.57	1	6	12
TotalSpent	200	13,229.15	7,492.80	1,086.01	12,493.42	41,727.46
Visits	200	8.14	2.86	2	8	16
SupportCalls	200	1.85	1.37	0	2	6
TenureMonths	200	30.33	17.37	1	30	60

4. Part A: Fundamentals

4.1 What is Data Analysis?

Data analysis is the process of inspecting, cleaning, transforming and modelling data to discover useful information, draw conclusions and support decisions. It ranges from descriptive analysis (what happened) and diagnostic analysis (why it happened) to predictive analysis (what will happen) and prescriptive analysis (what should be done).

4.2 Steps to Plan a Data Science Project

- **Define the problem:** state the business question and how success is measured.
- **Collect data:** CSV files, JSON, databases and APIs.
- **Understand the data:** structure, types, quality and profiling.
- **Clean and preprocess:** missing values, duplicates, data types, outliers, irrelevant columns.
- **Explore (EDA):** univariate, bivariate and multivariate analysis with visualizations.
- **Engineer features:** encode categoricals, scale numerics, create new variables.
- **Model:** select algorithms, train and tune them.
- **Evaluate:** accuracy, precision, recall, F1 and ROC-AUC.
- **Deploy and monitor:** serve the model and watch for drift.
- **Communicate:** report findings and document the work.

4.3 Machine Learning Problem Statement

Problem type: supervised binary classification. **Statement:** given a customer's demographics and purchase behaviour (Age, Gender, Income, Purchases, TotalSpent, Visits, SupportCalls, City, TenureMonths), predict whether the customer will churn (Yes) or stay (No). **Target:** `Churn`, with 39% positive cases, so the classes are moderately imbalanced and metrics such as precision, recall, F1 and ROC-AUC should be reported alongside accuracy.

4.4 Tensors

A tensor is an n-dimensional array of numbers and the basic data container in machine learning. Its **rank** is the number of axes and its **shape** gives the size along each axis. Tabular data is a 2-D tensor (rows x features), a colour image is 3-D (height x width x channels), and a batch of images is 4-D. Libraries such as TensorFlow and PyTorch build on the same idea and add GPU acceleration and automatic differentiation.

Rank	Name	Notebook example	Shape
0	Scalar	np.array(10)	()
1	Vector	np.array([1,2,3])	(3,)
2	Matrix	np.array([[1,2,3],[4,5,6]])	(2, 3)
3	3-D tensor	np.array([[[1,2],[3,4]],[[5,6],[7,8]]])	(2, 2, 2)

```
import numpy as np
scaler = np.array(10);  print(scaler.shape)        # ()
vector = np.array([1, 2, 3]);  print(vector.shape)  # (3,)
mat = np.array([[1, 2, 3], [4, 5, 6]]);  print(mat.shape)   # (2, 3)
tensor_3d = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])
print(tensor_3d.shape)                              # (2, 2, 2)
```

5. Part B: Data Acquisition

Source	Method used
CSV	`pd.read_csv("customer_churn_200.csv", sep=",", encoding="utf-8", header=0)`
JSON	`pd.read_json("customer_churn_200.json")`
SQL (MySQL)	`mysql.connector.connect(host="localhost", user="root", database="customer_churn")` then `pd.read_sql_query("select * from customer_data", conn)`
REST API	`requests.get(url)` on a Postman mock endpoint, then `pd.DataFrame(response.json()["data"])`

All four sources returned customer records with the same kind of columns. The CSV DataFrame (`df_csv`) was used for the rest of the analysis.

6. Part C: Data Understanding and Cleaning

6.1 Initial exploration

`.head()`, `.info()`, `.describe().T`, `.isnull().sum()`, `.duplicated().sum()` and `.shape` were used. The dataset has 200 rows and 11 columns (4 integer, 3 float, 4 object dtypes), occupying 17.3 KB.

Check	Result
Shape	(200, 11)
Duplicate rows	0
Missing: Age	6 (3.0%)
Missing: Gender	3 (1.5%)
Missing: Income	5 (2.5%)
Missing: all other columns	0

6.2 Cleaning steps applied

Issue	Treatment	Reason
Duplicates	`drop_duplicates()` (none found; run for practice)	Guarantees one row per record
Age: 6 missing	Filled with the mean (42.14), then cast to integer	Roughly symmetric distribution (skewness -0.14); age should be whole years
Gender: 3 missing	Filled with the mode; whitespace stripped and capitalised	Standard fix for categorical gaps; removes case/space variants
Income: 5 missing	Filled with the mean (70,696.35)	Roughly symmetric distribution (skewness -0.09)

```
df_csv["Age"] = df_csv["Age"].fillna(df_csv["Age"].mean())
df_csv["Age"] = df_csv["Age"].astype(int)
df_csv["Gender"] = df_csv["Gender"].fillna(df_csv["Gender"].mode()[0])
df_csv["Gender"] = df_csv["Gender"].str.strip().str.capitalize()
df_csv["Income"] = df_csv["Income"].fillna(df_csv["Income"].mean())
```

After cleaning, the profiling report confirms 0 missing cells and 0 duplicate rows.

7. Part D: Exploratory Data Analysis

7.1 Univariate analysis

Histograms, KDE plots and box plots were drawn for Age, Income and Purchases.

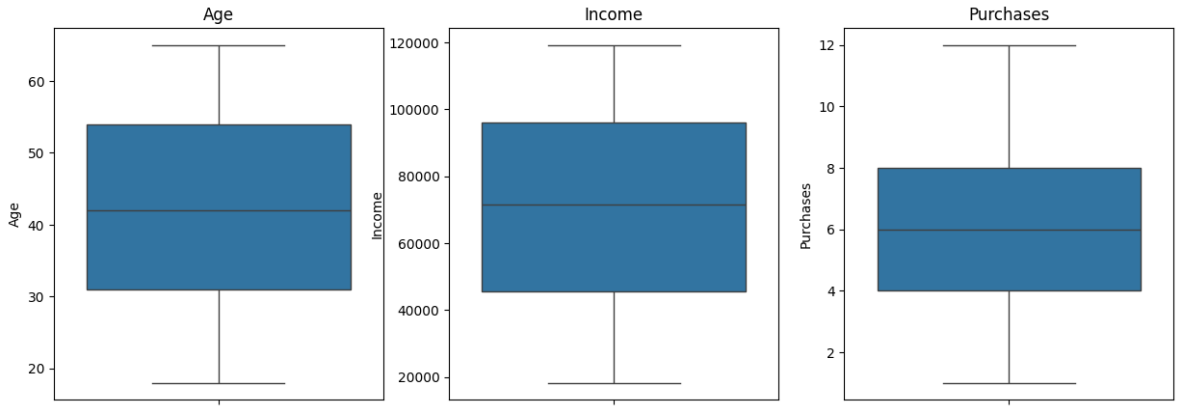


Figure 1: Box plots of Age, Income and Purchases

- **Age:** median 42, IQR 31 to 54, range 18 to 65; near-symmetric and no outliers.
- **Income:** median about 71,600, IQR about 45,700 to 96,200; evenly spread and no outliers.
- **Purchases:** median 6, IQR 4 to 8, range 1 to 12; slightly right-skewed (0.12) and no outliers.

7.2 Bivariate analysis

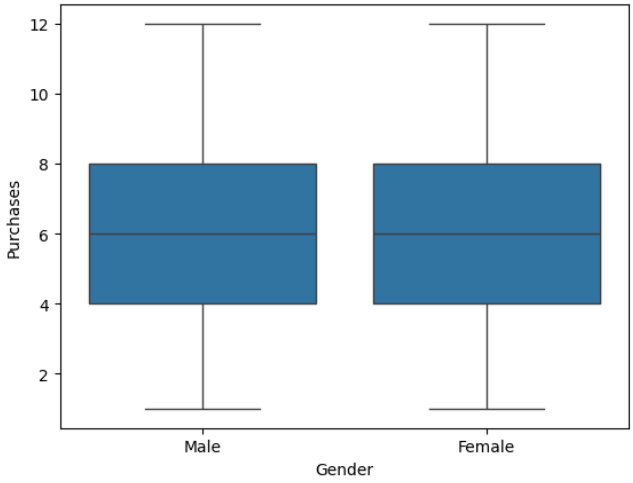


Figure 2: Purchases by gender

Gender vs Purchases: both genders have the same median (6) and the same interquartile range (4 to 8), so gender has no visible influence on purchase volume.

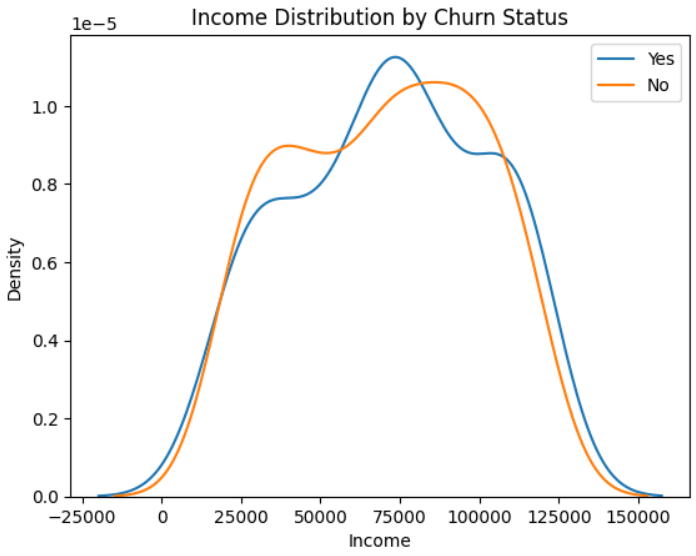


Figure 3: Income distribution by churn status

Income vs Churn: the two density curves overlap heavily. Churners are slightly more concentrated around 70,000, while retained customers show mild peaks near 40,000 and 85,000. There is no clear separation, so income alone is a weak predictor of churn.

7.3 Multivariate analysis

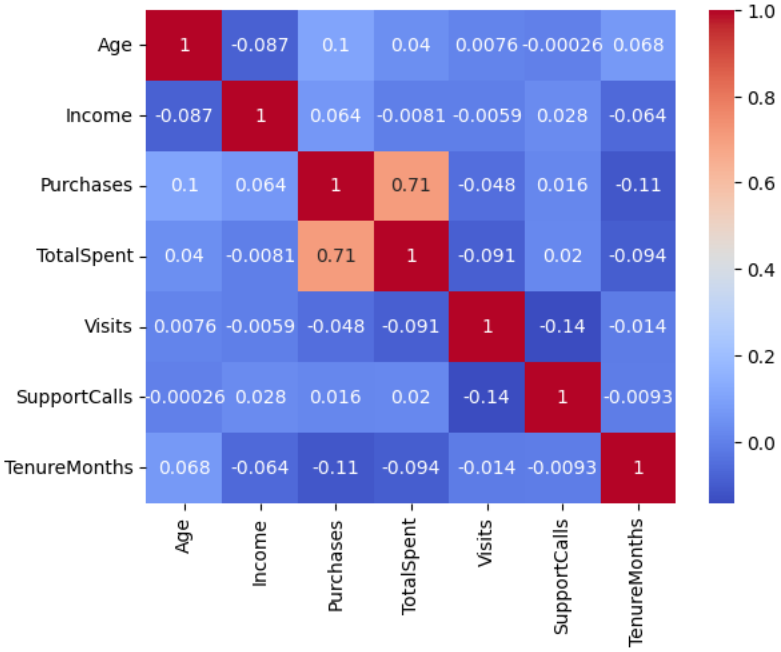


Figure 4: Correlation heatmap of numerical variables

Correlation heatmap: Purchases and TotalSpent are strongly correlated (r = 0.71). All other pairs are weak (absolute r below 0.15); the largest is Visits vs SupportCalls (r = -0.14). A pair plot of Age, Income, Purchases and TotalSpent was also generated to inspect feature interactions and shows the Purchases-TotalSpent relationship as the only visible trend.

8. Part E: Data Profiling

The profiling report was generated with YData Profiling v4.18.4 (the successor of pandas-profiling) in 12.65 seconds and saved as `ProfileReport.html`.

```
from ydata_profiling import ProfileReport
profile = ProfileReport(df_csv)
profile.to_file(output_file="ProfileReport.html")
```

Metric	Value
Variables	11 (1 text, 7 numeric, 2 categorical, 1 boolean)
Observations	200
Missing cells	0 (0.0%)
Duplicate rows	0 (0.0%)
Memory	17.3 KiB
Alerts	5

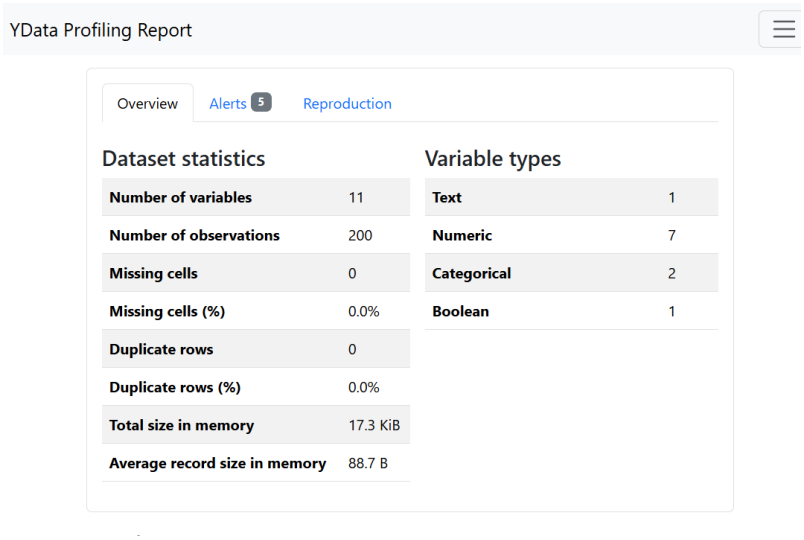


Figure 5: Profiling report overview

8.1 Alerts

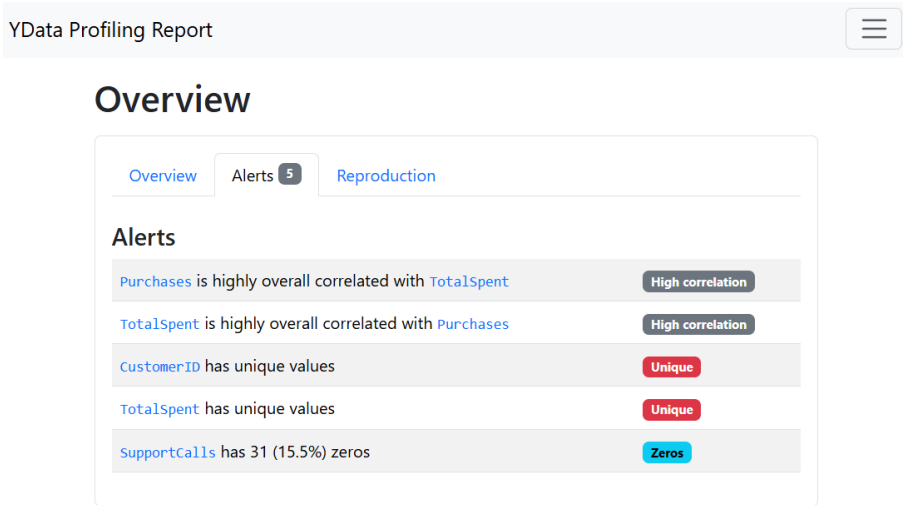


Figure 6: The five alerts raised by the report

Alert	Meaning and action
Purchases / TotalSpent high correlation	r = 0.72 (report) - keep one, or create average spend per purchase
CustomerID unique	Identifier only - drop before modelling
TotalSpent unique	Continuous value with no repeats - expected, no action needed
SupportCalls 31 zeros (15.5%)	Customers who never called support - keep, or add a "no support calls" flag

8.2 Variable detail: Income

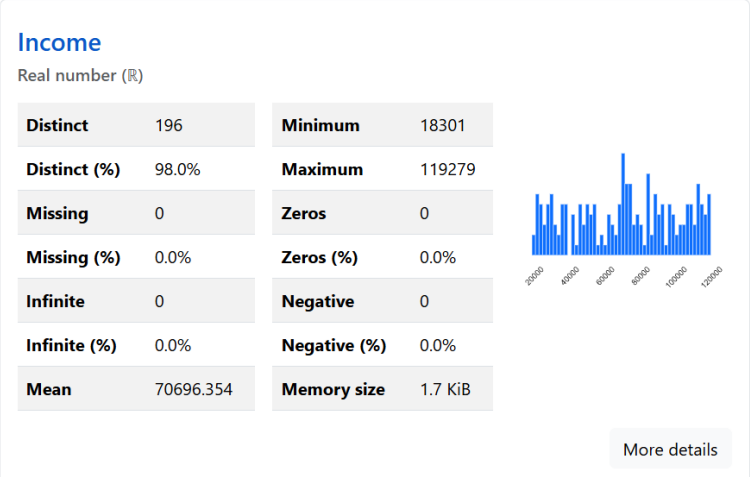


Figure 7: Income variable summary from the report

Income has 196 distinct values (98%), no zeros or negatives, and ranges from 18,301 to 119,279. The five imputed rows share the mean value 70,696.35, which explains why that value appears 5 times (2.5%) in the report.

9. Key Findings

- The dataset is clean and small: 14 missing cells out of 2,200 (0.6%) and no duplicates.
- 39% of customers churned, so the target is moderately imbalanced.
- Purchases and TotalSpent overlap strongly (r = 0.71); keep one or combine them.
- Income and gender show almost no relationship with purchases or churn; predictive signal is likely to come from tenure, support calls, visits and city, or from combinations of features.
- CustomerID is an identifier and should be excluded from modelling.

10. Suggested Next Steps (Feature Engineering)

Step	How	Why
Encode target	Churn: Yes -> 1, No -> 0	Models need numeric labels
Encode categoricals	One-hot encode Gender and City	Models need numeric input
Drop identifier	Remove CustomerID	No predictive value
Create ratio	AvgSpend = TotalSpent / Purchases	Removes overlap between two correlated columns
Scale numerics	StandardScaler on Age, Income, TotalSpent etc.	Needed for distance-based and linear models
Split and model	Stratified train/test split; logistic regression or random forest	Keeps churn ratio equal in both sets

11. Notes for Improving the Notebook

- **Cell order:** the `astype(int)` cell for Age sits before the cell that fills missing Age values, and astype(int) fails while NaN values exist. Run the fill first, then the cast, or reorder the cells. Use Restart & Run All before submitting.
- **File paths:** the notebook uses absolute Windows paths. Use relative paths so anyone can run it from the repository.
- **API comment:** the heading says "random user api" but the URL is a Postman mock products endpoint. Update the label to match.
- **Interpretations:** the brief asks for short interpretations under each chart. Add a markdown cell under each result (the wording in Section 7 can be reused).
- **Gender vs Purchases bar chart:** `plt.bar` on raw rows draws one bar per customer. A grouped mean, such as `df_csv.groupby("Gender")["Purchases"].mean().plot.bar()`, is clearer.
- **Irrelevant columns:** the brief asks to drop irrelevant columns; add `df_csv.drop(columns=["CustomerID"])` when preparing for modelling.

12. Submission Checklist

Requirement	Status
Jupyter Notebook (main.ipynb) with code and outputs	Ready
Profiling report (ProfileReport.html)	Ready
Labelled charts with short interpretations	Charts ready; add interpretations to the notebook
Screenshots of profiling report	Ready (screenshots folder)
PDF explaining theory concepts (this document)	Ready
README.md	Ready
GitHub repository with all files	Upload the files listed in the README structure

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